CAS SciFinder

Task History

Initiating Search

November 6, 2025, 4:52 PM

Search:

CAS Lexicon Search:

Sodium borohydride - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (67,611)	**References**	View Results
Exported: Retrieved Related Reaction Results + Filters (1,155) Filtered By: Yield: **90-100%, 80-89%** Language: **English** Publication Year: **2015 to 2025**	**Reactions**	View Results

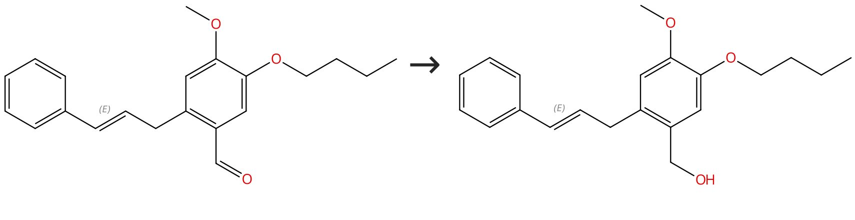
CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **100%**



Supplier (1)

31-513-CAS-22175942

Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium borohydride
Solvents: Methanol; 30 min, 25 °C

Experimental Protocols

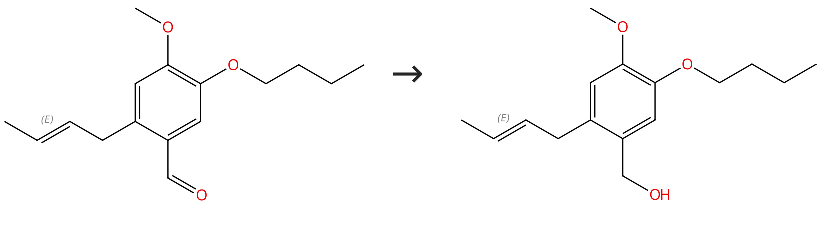
mCPBA-Mediated Intramolecular Oxidative Annulation of ortho-Crotyl or Cinnamyl Arylald ehydes: Synthesis of Benzofused Five-, Six-, and Seven-Membered Oxacycles

By: Chang, Meng-Yang; et al

Journal of Organic Chemistry (2018), 83(22), 14110-14119.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **100%**



31-513-CAS-19377685

Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium borohydride
Solvents: Methanol; 30 min, 25 °C

Experimental Protocols

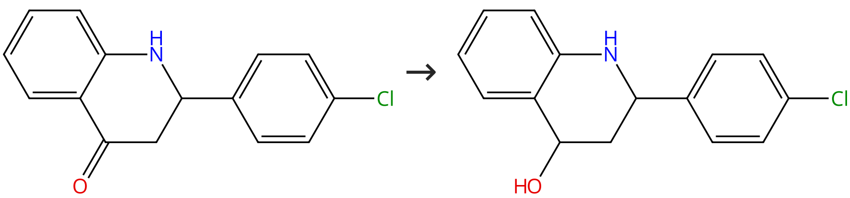
mCPBA-Mediated Intramolecular Oxidative Annulation of ortho-Crotyl or Cinnamyl Arylald ehydes: Synthesis of Benzofused Five-, Six-, and Seven-Membered Oxacycles

By: Chang, Meng-Yang; et al

Journal of Organic Chemistry (2018), 83(22), 14110-14119.

Scheme 3 (1 Reaction)

Steps: **1** Yield: **100%**

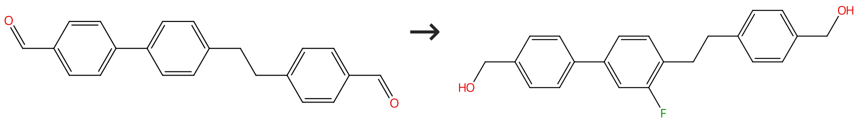


Suppliers (23)

Suppliers (2)

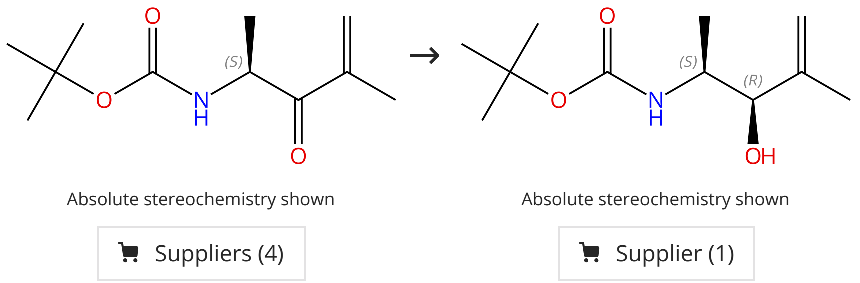
31-614-CAS-40418545	Steps: **1** Yield: **100%**	Using Quinolin-4-Ones as Convenient Common Precursors for a Metal-Free Total Synthesis of Both Dubamine and Graveoline Alkaloids and Diverse Structural Analogues
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol; 1 - 2 h, rt	By: Abonia, Rodrigo; et al
Experimental Protocols		Molecules (2024), 29(9), 1959.

Scheme 4 (1 Reaction) Steps: **1** Yield: **100%**



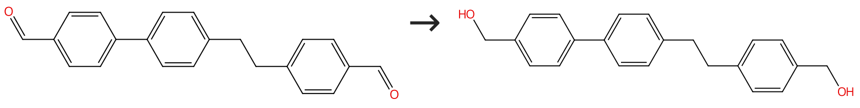
31-084-CAS-17674676	Steps: **1** Yield: **100%**	Liquid crystal compound having alkenyl on both ends, liquid crystal composition and liquid crystal display device
1.1	**Reagents:** Sodium borohydride **Solvents:** Tetrahydrofuran; < 5 °C; 30 min, 20 °C → 30 °C	Assignees: JNC Corporation; JNC Petrochemical Corporation
1.2	**Reagents:** Ammonium chloride **Solvents:** Water; < 10 °C	United States, US20170051206 A1 2017-02-23
		PatentPak available

Scheme 5 (1 Reaction) Steps: **1** Yield: **100%**



31-513-CAS-10860157	Steps: **1** Yield: **100%**	Systematic Analyses of Substrate Preferences of 20S Proteasomes Using Peptidic Epoxyketone Inhibitors
1.1	**Reagents:** Cerium trichloride heptahydrate **Solvents:** Methanol; rt → 0 °C	By: Huber, Eva M.; et al
1.2	**Reagents:** Sodium borohydride; 10 min, 0 °C; 30 min, 0 °C	Journal of the American Chemical Society (2015), 137(24), 7835-7842.
1.3	**Reagents:** Acetic acid; 15 min, 0 °C	
Experimental Protocols		

Scheme 6 (1 Reaction) Steps: **1** Yield: **100%**



31-513-CAS-17674668	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Tetrahydrofuran; 5 °C; 3 h, 20 °C → 30 °C
1.2	**Reagents:** Ammonium chloride **Solvents:** Water; 10 °C

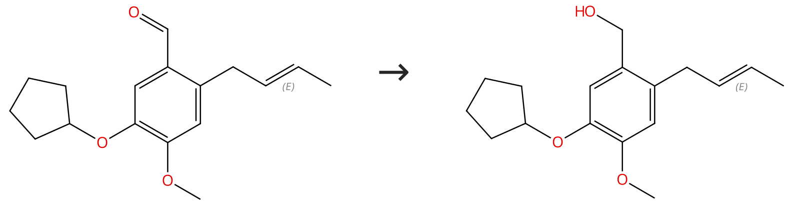
Liquid crystal compound having alkenyl on both ends, liquid crystal composition and liquid crystal display device

Assignees: JNC Corporation; JNC Petrochemical Corporation

United States, US20170051206 A1 2017-02-23

PatentPak available

Scheme 7 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown   Double bond geometry shown

31-513-CAS-19377684	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol; 30 min, 25 °C

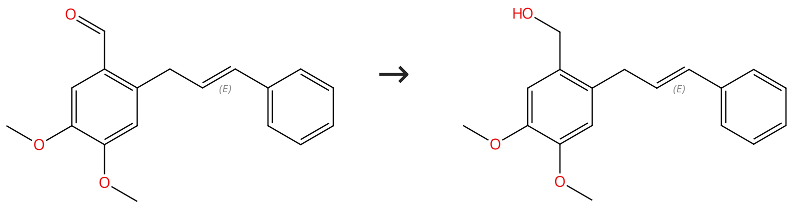
Experimental Protocols

mCPBA-Mediated Intramolecular Oxidative Annulation of ortho-Crotyl or Cinnamyl Arylaldehydes: Synthesis of Benzofused Five-, Six-, and Seven-Membered Oxacycles

By: Chang, Meng-Yang; et al

Journal of Organic Chemistry (2018), 83(22), 14110-14119.

Scheme 8 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown   Double bond geometry shown

Supplier (1)

31-513-CAS-19377678	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol; 30 min, 25 °C

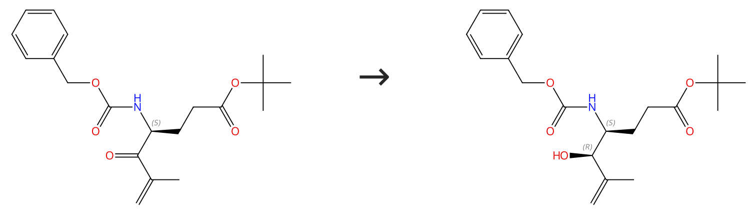
Experimental Protocols

mCPBA-Mediated Intramolecular Oxidative Annulation of ortho-Crotyl or Cinnamyl Arylaldehydes: Synthesis of Benzofused Five-, Six-, and Seven-Membered Oxacycles

By: Chang, Meng-Yang; et al

Journal of Organic Chemistry (2018), 83(22), 14110-14119.

Scheme 9 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown   Absolute stereochemistry shown

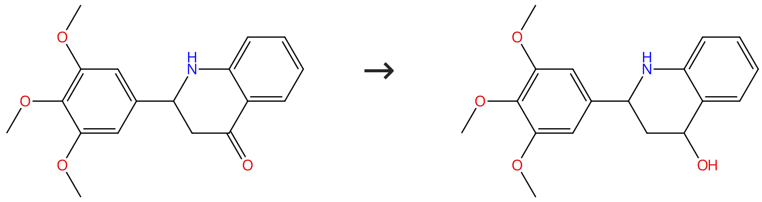
31-513-CAS-12641938	Steps: **1** Yield: **100%**
1.1	**Reagents:** Cerium trichloride heptahydrate **Solvents:** Methanol; rt → 0 °C
1.2	**Reagents:** Sodium borohydride; 10 min, 0 °C; 30 min, 0 °C
1.3	**Reagents:** Acetic acid; 15 min, 0 °C
Experimental Protocols	

Systematic Analyses of Substrate Preferences of 20S Proteasomes Using Peptidic Epoxyketone Inhibitors

By: Huber, Eva M.; et al

Journal of the American Chemical Society (2015), 137(24), 7835-7842.

Scheme 10 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (7) Supplier (1)

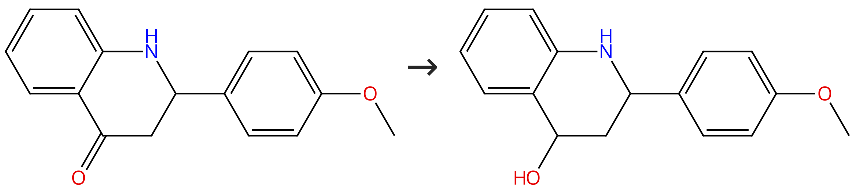
31-614-CAS-40418546	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol; 1 - 2 h, rt
Experimental Protocols	

Using Quinolin-4-Ones as Convenient Common Precursors for a Metal-Free Total Synthesis of Both Dubamine and Graveoline Alkaloids and Diverse Structural Analogues

By: Abonia, Rodrigo; et al

Molecules (2024), 29(9), 1959.

Scheme 11 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (16) Suppliers (2)

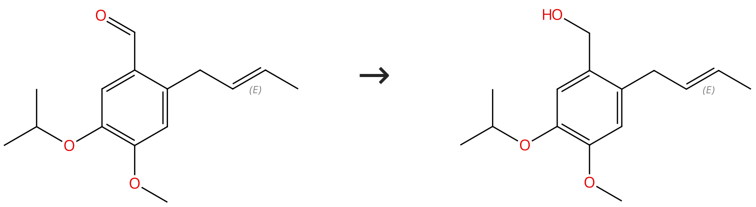
31-614-CAS-40418543	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol; 1 - 2 h, rt
Experimental Protocols	

Using Quinolin-4-Ones as Convenient Common Precursors for a Metal-Free Total Synthesis of Both Dubamine and Graveoline Alkaloids and Diverse Structural Analogues

By: Abonia, Rodrigo; et al

Molecules (2024), 29(9), 1959.

Scheme 12 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown Double bond geometry shown

31-513-CAS-19377686 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium borohydride
Solvents: Methanol; 30 min, 25 °C

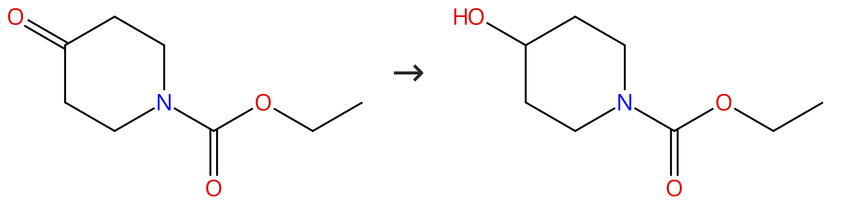
Experimental Protocols

mCPBA-Mediated Intramolecular Oxidative Annulation of ortho-Crotyl or Cinnamyl Arylaldehydes: Synthesis of Benzofused Five-, Six-, and Seven-Membered Oxacycles

By: Chang, Meng-Yang; et al

Journal of Organic Chemistry (2018), 83(22), 14110-14119.

Scheme 13 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (76) Suppliers (74)

31-513-CAS-17848736 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium borohydride
Solvents: Methanol; 2 h, 0 °C

1.2 **Reagents:** Ammonium chloride
Solvents: Water

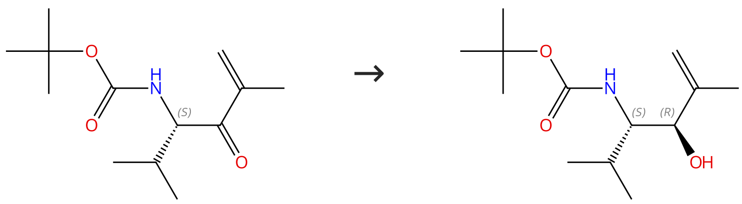
Preparation and antibacterial activity of gold complexes with phosphine and thiolate compounds

Assignee: Auspherix Limited

World Intellectual Property Organization, WO2017093543 A2 2017-06-08

PatentPak available

Scheme 14 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown Absolute stereochemistry shown

Suppliers (4)

31-513-CAS-12975084 Steps: **1** Yield: **100%**

1.1 **Reagents:** Cerium trichloride heptahydrate
Solvents: Methanol; rt → 0 °C

1.2 **Reagents:** Sodium borohydride; 10 min, 0 °C; 30 min, 0 °C

1.3 **Reagents:** Acetic acid; 15 min, 0 °C

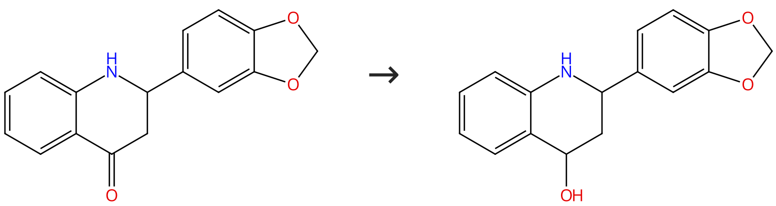
Experimental Protocols

Systematic Analyses of Substrate Preferences of 20S Proteasomes Using Peptidic Epoxyketone Inhibitors

By: Huber, Eva M.; et al

Journal of the American Chemical Society (2015), 137(24), 7835-7842.

Scheme 15 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (15)

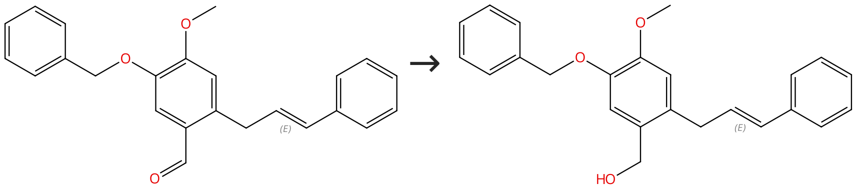
31-614-CAS-40418560	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol; 1 - 2 h, rt
Experimental Protocols	

Using Quinolin-4-Ones as Convenient Common Precursors for a Metal-Free Total Synthesis of Both Dubamine and Graveoline Alkaloids and Diverse Structural Analogues

By: Abonia, Rodrigo; et al

Molecules (2024), 29(9), 1959.

Scheme 16 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown Double bond geometry shown

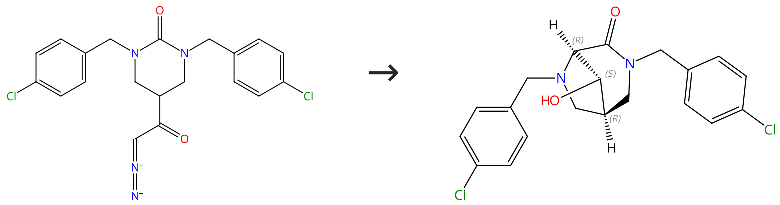
31-513-CAS-19377679	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol; 30 min, 25 °C
Experimental Protocols	

mCPBA-Mediated Intramolecular Oxidative Annulation of ortho-Crotyl or Cinnamyl Arylaldehydes: Synthesis of Benzofused Five-, Six-, and Seven-Membered Oxacycles

By: Chang, Meng-Yang; et al

Journal of Organic Chemistry (2018), 83(22), 14110-14119.

Scheme 17 (1 Reaction) Steps: **1** Yield: **100%**



Relative stereochemistry shown

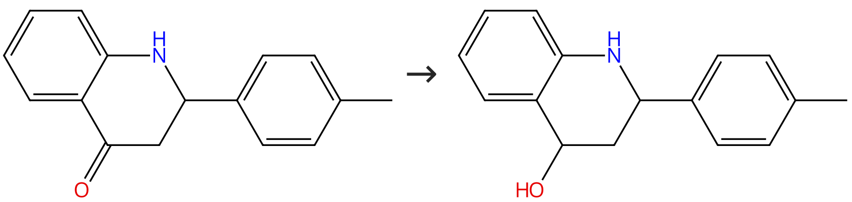
31-513-CAS-20125582	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Rhodium, tetrakis[μ-(2,2-dimethylpropanamidato-κ*N*:κ*O*)]di-, (*Rh*-*Rh*) **Solvents:** Dichloromethane, 1,4-Dioxane; 1 min, 40 °C; 30 min, 40 °C
1.2	**Reagents:** Lithium tri-*tert*-butoxyaluminum hydride **Solvents:** Tetrahydrofuran; 3 h, -78 °C
1.3	**Reagents:** Potassium sodium tartrate **Solvents:** Water

Urea insertion reaction of rhodium-carbenoid

By: Hashimoto, Yoshinori; et al

Chemical & Pharmaceutical Bulletin (2018), 66(11), 1041-1047.

Scheme 18 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (6) Suppliers (2)

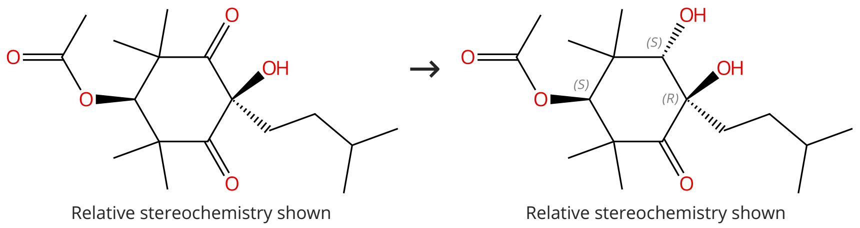
31-614-CAS-40418550	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol; 1 - 2 h, rt
Experimental Protocols	

Using Quinolin-4-Ones as Convenient Common Precursors for a Metal-Free Total Synthesis of Both Dubamine and Graveoline Alkaloids and Diverse Structural Analogues

By: Abonia, Rodrigo; et al

Molecules (2024), 29(9), 1959.

Scheme 19 (1 Reaction) Steps: **1** Yield: **100%**



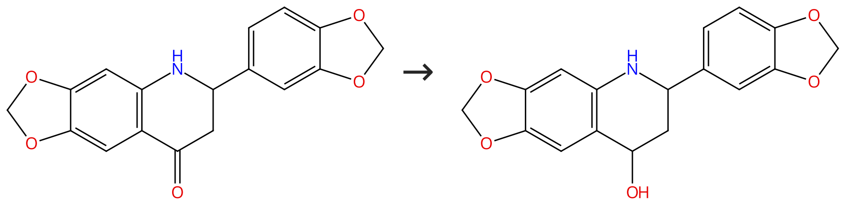
31-513-CAS-19246155	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium triacetoxyborohydride **Solvents:** Tetrahydrofuran; 18 h, rt
1.2	**Reagents:** Sodium chloride **Solvents:** Water; rt
Experimental Protocols	

Improved total synthesis of (±)-Tetragocarbone A

By: Nishimura, Eiji; et al

Tetrahedron (2018), 74(21), 2664-2668.

Scheme 20 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (4)

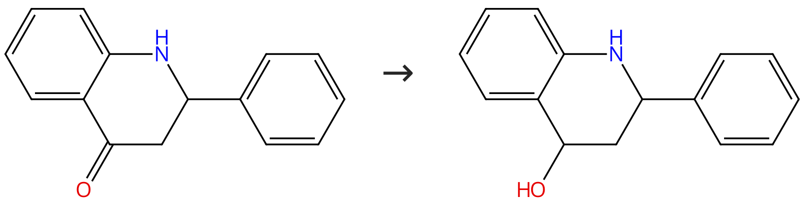
31-614-CAS-40418542	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol; 1 - 2 h, rt
Experimental Protocols	

Using Quinolin-4-Ones as Convenient Common Precursors for a Metal-Free Total Synthesis of Both Dubamine and Graveoline Alkaloids and Diverse Structural Analogues

By: Abonia, Rodrigo; et al

Molecules (2024), 29(9), 1959.

Scheme 21 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (48) Suppliers (2)

31-614-CAS-40418548 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium borohydride
Solvents: Methanol; 1 - 2 h, rt

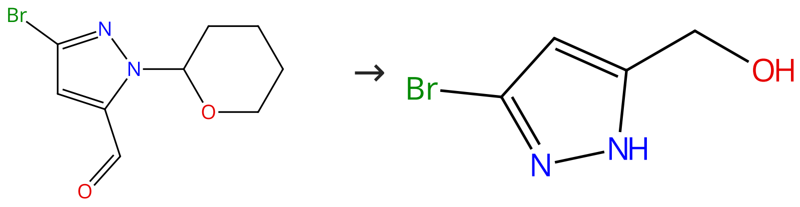
Experimental Protocols

Using Quinolin-4-Ones as Convenient Common Precursors for a Metal-Free Total Synthesis of Both Dubamine and Graveoline Alkaloids and Diverse Structural Analogues

By: Abonia, Rodrigo; et al

Molecules (2024), 29(9), 1959.

Scheme 22 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (43) Suppliers (4)

31-614-CAS-34716841 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium borohydride
Solvents: Methanol; overnight, rt

1.2 **Reagents:** Hydrochloric acid
Solvents: 1,4-Dioxane; 1 h, rt

1.3 **Reagents:** Sodium bicarbonate
Solvents: Water; neutralized

Experimental Protocols

JDQ443, a structurally novel, pyrazole-based, covalent inhibitor of KRAS^{G12C} for the treatment of solid tumors

By: Lorthiois, Edwige; et al

Journal of Medicinal Chemistry (2022), 65(24), 16173-16203.

Scheme 23 (1 Reaction) Steps: **1** Yield: **100%**

Absolute stereochemistry shown Absolute stereochemistry shown

Suppliers (2)

31-513-CAS-1671023 Steps: **1** Yield: **100%**

1.1 **Reagents:** Cerium trichloride heptahydrate
Solvents: Methanol; rt → 0 °C

1.2 **Reagents:** Sodium borohydride; 10 min, 0 °C; 30 min, 0 °C

1.3 **Reagents:** Acetic acid; 15 min, 0 °C

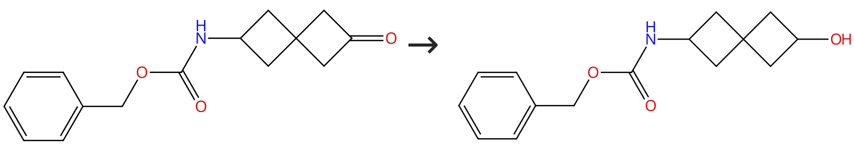
Experimental Protocols

Systematic Analyses of Substrate Preferences of 20S Proteasomes Using Peptidic Epoxyketone Inhibitors

By: Huber, Eva M.; et al

Journal of the American Chemical Society (2015), 137(24), 7835-7842.

Scheme 24 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (42) Suppliers (19)

31-614-CAS-40788180	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol, Tetrahydrofuran; 30 min, 0 °C; 0 °C → rt; 30 min, rt
1.2	**Reagents:** Ammonium chloride **Solvents:** Water

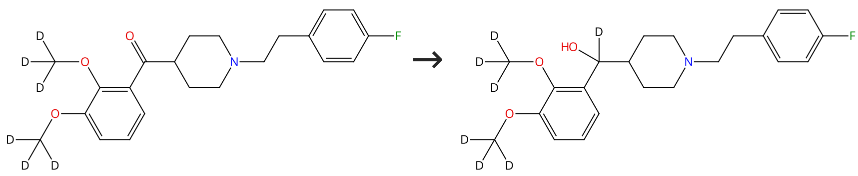
Spiroheptanyl hydantoins as rock inhibitors

Assignee: Bristol-Myers Squibb Company

United States, US20240092755 A1 2024-03-21

PatentPak available

Scheme 25 (1 Reaction) Steps: **1** Yield: **100%**



31-513-CAS-23335962	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borodeuteride **Solvents:** Methanol-*d*; 0 °C → rt; 16 h, rt

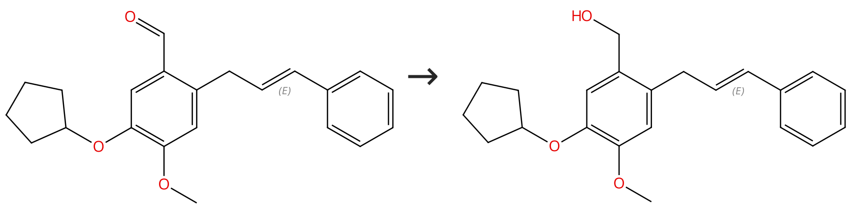
Deuterated forms and derivatives of volinanserin

Assignee: Concert Pharmaceuticals, Inc.

World Intellectual Property Organization, WO2020132461 A1 2020-06-25

PatentPak available

Scheme 26 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown   Double bond geometry shown

31-513-CAS-19377680	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol; 30 min, 25 °C

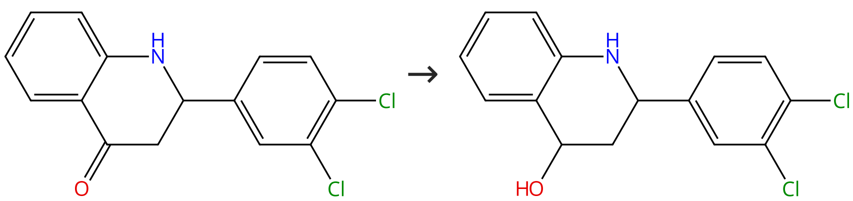
Experimental Protocols

mCPBA-Mediated Intramolecular Oxidative Annulation of ortho-Crotyl or Cinnamyl Arylaldehydes: Synthesis of Benzofused Five-, Six-, and Seven-Membered Oxacycles

By: Chang, Meng-Yang; et al

Journal of Organic Chemistry (2018), 83(22), 14110-14119.

Scheme 27 (1 Reaction) Steps: **1** Yield: **100%**



Supplier (1)

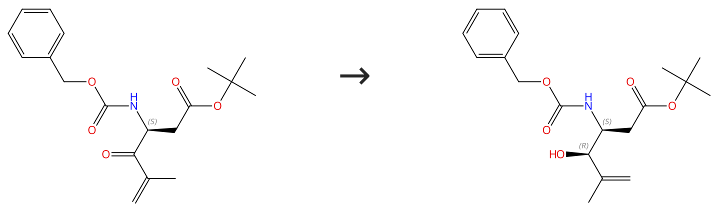
31-614-CAS-40418553	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol; 1 - 2 h, rt
Experimental Protocols	

Using Quinolin-4-Ones as Convenient Common Precursors for a Metal-Free Total Synthesis of Both Dubamine and Graveoline Alkaloids and Diverse Structural Analogues

By: Abonia, Rodrigo; et al

Molecules (2024), 29(9), 1959.

Scheme 28 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown

Absolute stereochemistry shown

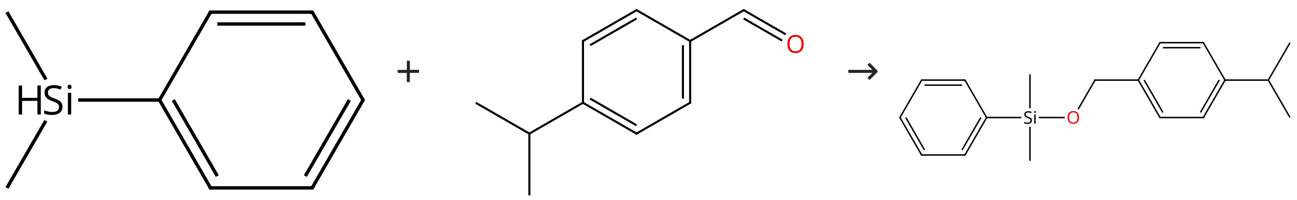
31-513-CAS-6217266	Steps: **1** Yield: **100%**
1.1	**Reagents:** Cerium trichloride heptahydrate **Solvents:** Methanol; rt → 0 °C
1.2	**Reagents:** Sodium borohydride; 10 min, 0 °C; 30 min, 0 °C
1.3	**Reagents:** Acetic acid; 15 min, 0 °C
Experimental Protocols	

Systematic Analyses of Substrate Preferences of 20S Proteasomes Using Peptidic Epoxyketone Inhibitors

By: Huber, Eva M.; et al

Journal of the American Chemical Society (2015), 137(24), 7835-7842.

Scheme 29 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (65)

Suppliers (92)

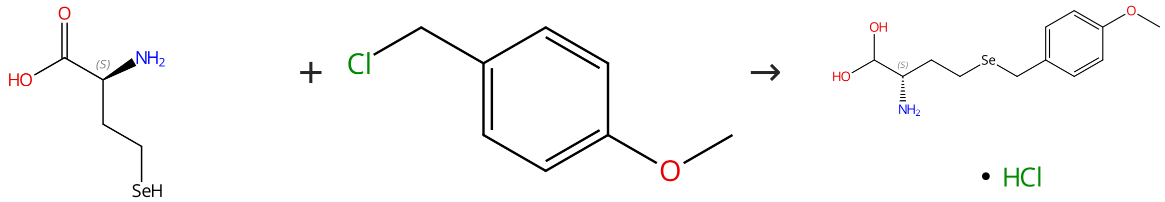
31-010-CAS-21238990	Steps: **1** Yield: **100%**
1.1	**Catalysts:** (3-Aminopropyl)triethoxysilane, Gold, Silica **Solvents:** Tetrahydrofuran; 24 h, 75 °C
Experimental Protocols	

Organosilane oxidation with a half million turnover number using fibrous nanosilica supported ultrasmall nanoparticles and pseudo-single atoms of gold

By: Dhiman, Mahak; et al

Journal of Materials Chemistry A: Materials for Energy and Sustainability (2017), 5(5), 1935-1940.

Scheme 30 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown

Suppliers (5)

Suppliers (70)

Absolute stereochemistry shown

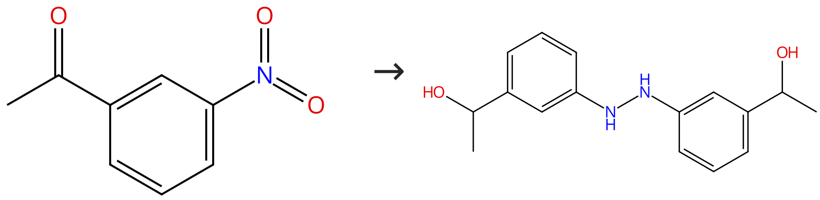
31-026-CAS-17972736	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium hydroxide, Sodium borohydride **Solvents:** Water; 15 min, rt → 0 °C
1.2	**Reagents:** Nitric acid; pH 6, 0 °C
1.3	1.5 h, 0 °C
1.4	**Reagents:** Hydrochloric acid **Solvents:** Water; pH 1 - 2
Experimental Protocols	

Revisiting ligation at selenomethionine: Insights into native chemical ligation at selenocysteine and homoselenocysteine

By: Dardashti, Rebecca Notis; et al

Bioorganic & Medicinal Chemistry (2017), 25(18), 4983-4989.

Scheme 31 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (84)

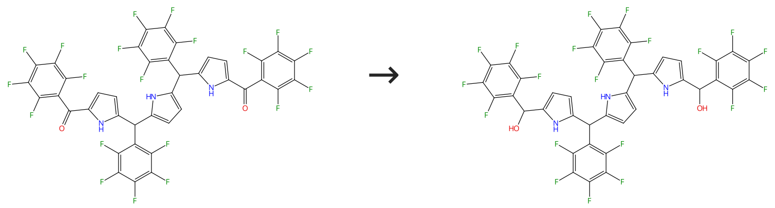
31-513-CAS-23074544	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Catalysts:** α-Phenyl-2-quinolinemethanol **Solvents:** Toluene, Tetrahydrofuran; 6 h, 110 °C
1.2	**Solvents:** Water

Organocatalytic Reduction of Nitroarenes with Phenyl(2-quinolyl)methanol

By: Giomi, Donatella; et al

ChemistrySelect (2020), 5(34), 10511-10515.

Scheme 32 (1 Reaction) Steps: **1** Yield: **100%**



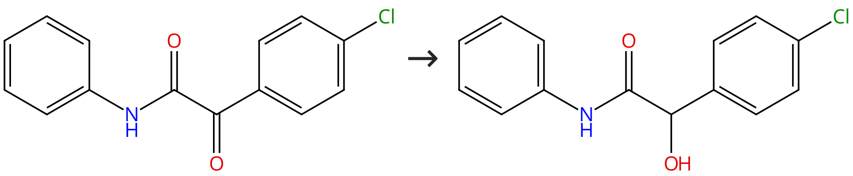
31-513-CAS-19037025	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol, Tetrahydrofuran; 1 h, 25 °C
1.2	**Solvents:** Water; 25 °C
Experimental Protocols	

Stable meso-Aryl β-Alkyl Hybrid Sapphyrin with a Warped π-Conjugation Circuit and Neo-Confused Sapphyrin-Silver(I) Complex

By: Mori, Daiki; et al

Chemistry - An Asian Journal (2018), 13(8), 934-938.

Scheme 33 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (4) Suppliers (18)

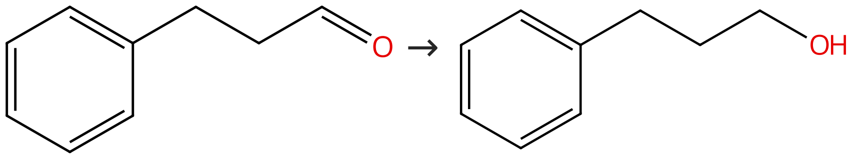
31-513-CAS-19122398	Steps: **1** Yield: **100%**
1.1	**Reagents:** 4-Methyl-α-oxo-*N*-phenylbenzeneacetamide **Catalysts:** Ceria **Solvents:** Methanol; 10 min, 45 °C
1.2	**Reagents:** Sodium borohydride; 30 min, rt
Experimental Protocols	

Nanoceria-catalyzed selective synthesis of α-hydroxy amides through the reduction of an unusual class of α-keto amides

By: Mishra, Ashish A.; et al

Asian Journal of Organic Chemistry (2018), 7(5), 922-931.

Scheme 34 (1 Reaction) — Steps: **1** Yield: **100%**



Suppliers (90) Suppliers (87)

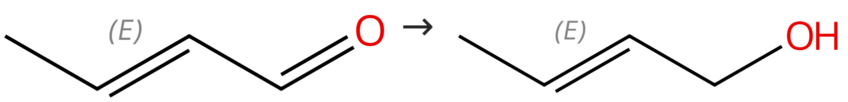
31-513-CAS-11456148	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Catalysts:** 1659248-89-1 **Solvents:** Methanol; 5 min, rt

Sodium borohydride reduction of aldehydes catalyzed by an oxovanadium(IV) Schiff base complex encapsulated in the nanocavity of zeolite-Y

By: Rayati, Saeed; et al

Inorganic Chemistry Communications (2015), 54, 38-40.

Scheme 35 (1 Reaction) — Steps: **1** Yield: **100%**



Double bond geometry shown Double bond geometry shown

Suppliers (32) Suppliers (28)

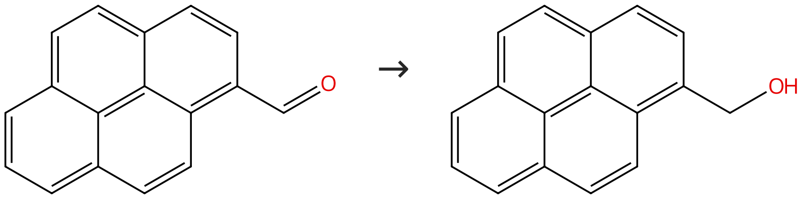
31-513-CAS-2632310	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Catalysts:** 1659248-89-1 **Solvents:** Methanol; 2 min, rt

Sodium borohydride reduction of aldehydes catalyzed by an oxovanadium(IV) Schiff base complex encapsulated in the nanocavity of zeolite-Y

By: Rayati, Saeed; et al

Inorganic Chemistry Communications (2015), 54, 38-40.

Scheme 36 (1 Reaction) — Steps: **1** Yield: **100%**



Suppliers (73) Suppliers (64)

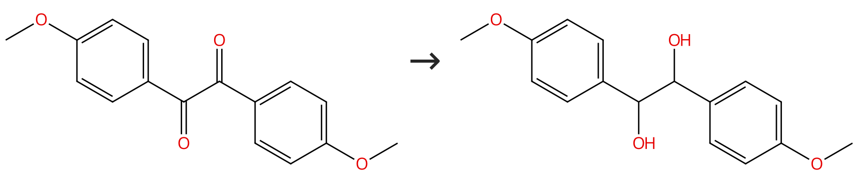
31-614-CAS-35789612	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Ethanol, Tetrahydrofuran; 0 °C; 16 h, rt
Experimental Protocols	

(1,6)Pyrenophanes containing crown ether moieties as fluorescence sensors for metal and ammonium ions. Formation of sandwich, dumbbell, and pseudorotaxane complexes

By: Maeda, Hajime; et al

New Journal of Chemistry (2023), 47(3), 1388-1400.

Scheme 37 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (75) Suppliers (5)

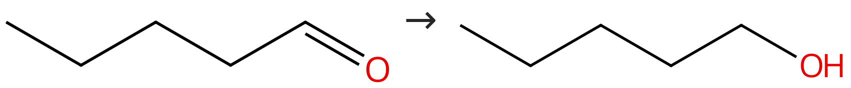
31-614-CAS-32459675	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium borohydride **Solvents:** Methanol, Tetrahydrofuran; 2 h, rt
Experimental Protocols	

Challenging Approach to the Development of Novel Estrogen Receptor Modulators Based on the Chemical Properties of Guaiazulene

By: Ohta, Kiminori; et al

International Journal of Molecular Sciences (2022), 23(3), 1113.

Scheme 38 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (57) Suppliers (79)

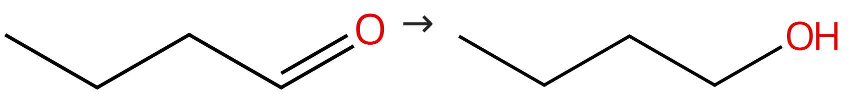
31-614-CAS-40233913	Steps: **1** Yield: **100%**
1.1	**Reagents:** Hydrogen **Catalysts:** Alumina, Nickel; 0.1 MPa, 40 °C
Experimental Protocols	

Colloidal and Nanosized Catalysts in Organic Synthesis: XXVI I.[1] Hydrogenation of Carbonyl Compounds in the Presence of Supported Nickel Catalysts

By: Razvalyaeva, A. V.; et al

Russian Journal of General Chemistry (2024), 94(2), 267-272.

Scheme 39 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (71) Suppliers (201)

31-614-CAS-40233911 Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Alumina, Nickel; 0.1 MPa, 20 °C

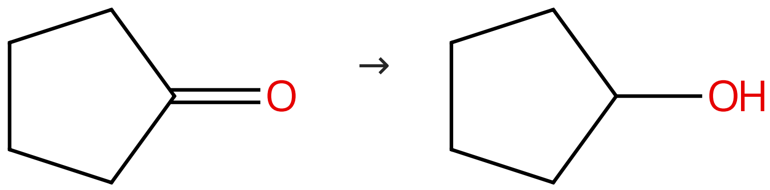
Experimental Protocols

Colloidal and Nanosized Catalysts in Organic Synthesis: XXVI I.[1] Hydrogenation of Carbonyl Compounds in the Presence of Supported Nickel Catalysts

By: Razvalyaeva, A. V.; et al

Russian Journal of General Chemistry (2024), 94(2), 267-272.

Scheme 40 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (78) Suppliers (59)

31-614-CAS-40233896 Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Alumina, Nickel; 0.1 MPa, 80 °C

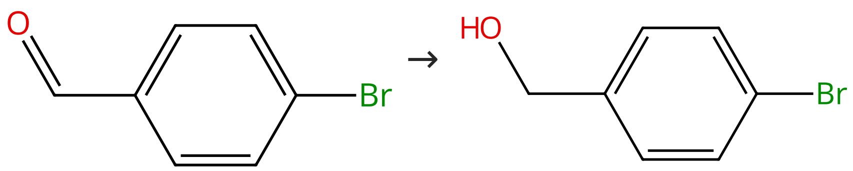
Experimental Protocols

Colloidal and Nanosized Catalysts in Organic Synthesis: XXVI I.[1] Hydrogenation of Carbonyl Compounds in the Presence of Supported Nickel Catalysts

By: Razvalyaeva, A. V.; et al

Russian Journal of General Chemistry (2024), 94(2), 267-272.

Scheme 41 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (91) Suppliers (88)

31-513-CAS-9675641 Steps: **1** Yield: **100%**

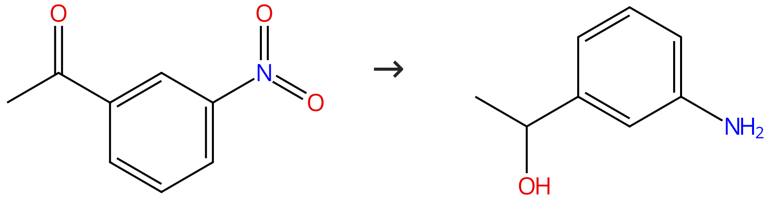
1.1 **Reagents:** Sodium borohydride
Catalysts: 1659248-89-1
Solvents: Methanol; 5 min, rt

Sodium borohydride reduction of aldehydes catalyzed by an oxovanadium(IV) Schiff base complex encapsulated in the nanocavity of zeolite-Y

By: Rayati, Saeed; et al

Inorganic Chemistry Communications (2015), 54, 38-40.

Scheme 42 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (84) Suppliers (63)

31-513-CAS-23143922 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium borohydride
Catalysts: Gold (Candida parapsilosis-supported)
Solvents: Water; 0 - 4 °C; 5 min, 4 °C; 5 h, rt

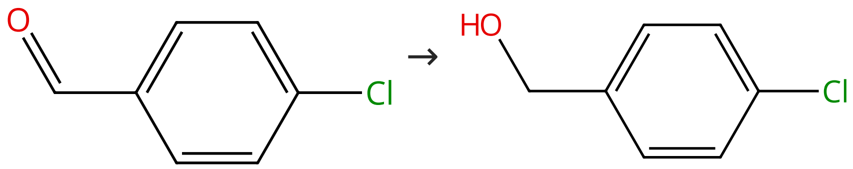
Experimental Protocols

Yeast supported gold nanoparticles: an efficient catalyst for the synthesis of commercially important aryl amines

By: Krishnan, Saravanan; et al

New Journal of Chemistry (2021), 45(4), 1915-1923.

Scheme 43 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (99) Suppliers (85)

31-513-CAS-5577507 Steps: **1** Yield: **100%**

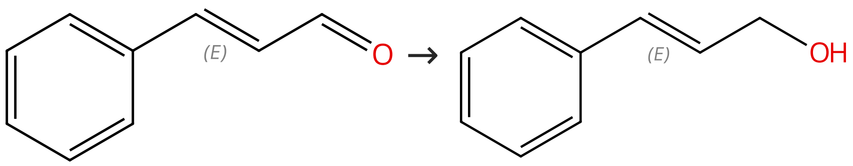
1.1 **Reagents:** Sodium borohydride
Catalysts: 1659248-89-1
Solvents: Methanol; 2 min, rt

Sodium borohydride reduction of aldehydes catalyzed by an oxovanadium(IV) Schiff base complex encapsulated in the nanocavity of zeolite-Y

By: Rayati, Saeed; et al

Inorganic Chemistry Communications (2015), 54, 38-40.

Scheme 44 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown Double bond geometry shown

Suppliers (109) Suppliers (66)

31-513-CAS-9303570 Steps: **1** Yield: **100%**

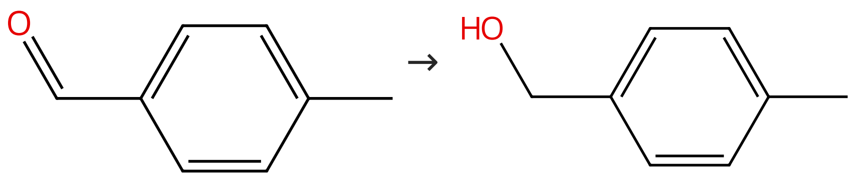
1.1 **Reagents:** Sodium borohydride
Catalysts: 1659248-89-1
Solvents: Methanol; 2 min, rt

Sodium borohydride reduction of aldehydes catalyzed by an oxovanadium(IV) Schiff base complex encapsulated in the nanocavity of zeolite-Y

By: Rayati, Saeed; et al

Inorganic Chemistry Communications (2015), 54, 38-40.

Scheme 45 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (104) Suppliers (79)

31-513-CAS-1496613 Steps: **1** Yield: **100%**

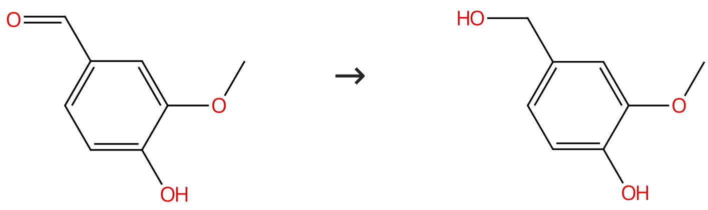
1.1 **Reagents:** Acetophenone, Sodium borohydride
Catalysts: 1659248-89-1
Solvents: Methanol; 2 min, rt

Sodium borohydride reduction of aldehydes catalyzed by an oxovanadium(IV) Schiff base complex encapsulated in the nanocavity of zeolite-Y

By: Rayati, Saeed; et al

Inorganic Chemistry Communications (2015), 54, 38-40.

Scheme 46 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (145)

Suppliers (98)

31-614-CAS-35597602 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium hydroxide, Sodium borohydride
Solvents: Ethanol, Water; 10 min, 0 °C; 10 min, rt; 0 °C

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; acidified, 0 °C

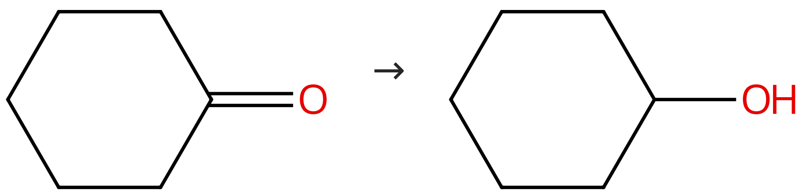
Experimental Protocols

Thermal annealing effects on the mechanical properties of biobased 3D printed thermosets

By: Cortes-Guzman, Karen P.; et al

ChemRxiv (2023), 1-20.

Scheme 47 (2 Reactions) Steps: **1** Yield: **100%**



Suppliers (102)

Suppliers (81)

31-614-CAS-40233899 Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrogen
Catalysts: Alumina, Nickel; 0.1 MPa, 80 °C

Experimental Protocols

Colloidal and Nanosized Catalysts in Organic Synthesis: XXVI I.[1] Hydrogenation of Carbonyl Compounds in the Presence of Supported Nickel Catalysts

By: Razvalyaeva, A. V.; et al

Russian Journal of General Chemistry (2024), 94(2), 267-272.

31-513-CAS-13363737 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium *tert*-butoxide, Hydrogen
Catalysts: (*OC*-6-15)-Dichloro[*N*-[2-(diphenylphosphino-κ*P*)ethyl]-2-pyridinemethanamine-κN^1,κN^2](triphenylphosphine)ruthenium
Solvents: Isopropanol; 18 h, 1 bar, rt

Experimental Protocols

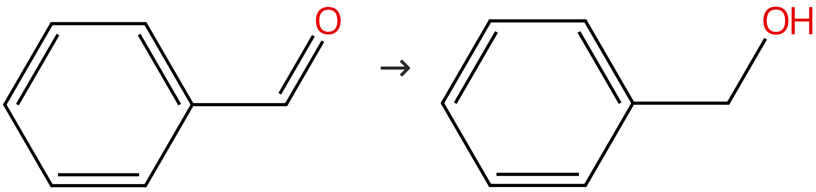
Probing the effect of heterocycle-bonding in PNX-type ruthenium pre-catalysts for reactions involving H_2

By: Xu, W.; et al

Dalton Transactions (2015), 44(38), 16785-16790.

Scheme 48 (2 Reactions)

Steps: **1** Yield: **100%**



Suppliers (69)

Suppliers (148)

31-614-CAS-31323085

Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium borohydride
Catalysts: Nickel acetate
Solvents: Water; 5 min, 50 °C

Experimental Protocols

A hassle-free and cost-effective transfer hydrogenation strategy for the chemoselective reduction of arylnitriles to primary amines through in situ-generated nickel (II) dihydride intermediate in water

By: Zeynizadeh, Behzad; et al

Journal of Molecular Structure (2022), 1255, 131926.

31-513-CAS-17352666

Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium benzoate, Sodium borohydride
Solvents: Water; 60 min, rt

1.2 **Solvents:** Water; rt

Experimental Protocols

Ultrasonic-promoted selective reduction of aldehydes vs. ketones by $NaBH_4/PhCO_2Na/H_2O$

By: Mirtaghizadeh, Mina; et al

Oriental Journal of Chemistry (2016), 32(3), 1539-1543.